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Preliminary paleomagnetic results from Jurassic rocks of the Santa Marta Massif, Colombia

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ABSTRACT

From the Sierra Nevada of Santa Marta at the northern terminus of the Andes, preliminary paleomagnetic results from ten sites in two formations characteristically have stable directions with northerly declinations and moderate positive inclinations. Most site mean directions are close in attitude to the present Earth's field and are not indicative of large tectonic rotations. An antipodally reversed direction is obtained at one site by thermal demagnetization, suggesting magnetization in an ancient paleofield nearly coaxial with the present field. Few cratonic Jurassic directions have been reported from South America, but at least one such study shows Jurassic paleofield directions close to the present field directions. In defining the Mesozoic polar wander path for South America, distinguishing Jurassic from present field directions will be a continuing challenge.

INTRODUCTION

As part of a paleomagnetic survey of Mesozoic rocks from South America (MacDonald and Opdyke, 1972; Opdyke and MacDonald, 1973; MacDonald and Opdyke, 1974), Jurassic red beds and volcanic rocks were sampled on the southeast flank of the Sierra Nevada of Santa Marta at the north end of the Andes. Preliminary results are summarized here for 75 samples from 11 sites in three Jurassic formations as mapped by Tschanz and others (1969). The red beds and volcanic rocks of this study are approximately coeval with those in the Perija mountains nearby to the east, from which paleomagnetic results are reported by Maze and Hargraves (1984).

GEOLOGIC SETTING

The Santa Marta massif lies at the northern end of the Colombian Andes. The oldest rocks here are 1,300-m.y.-old gneisses which are invaded by numerous batholiths of Jurassic and younger age (MacDonald and Hurley, 1969; Tschanz and others, 1969, 1974). These crystalline rocks are unconformably overlain and partly roofed by Jurassic terrestrial clastic rocks and silicic to basic volcanic rocks. Cretaceous marine strata unconformably overlie the Jurassic section.

This study presents preliminary paleomagnetic results for Jurassic red beds and volcanic rocks along the southeast flank of the massif (Fig. 1). In the region sampled, the prevailing dip is southeasterly, gentle to moderate. Many Mesozoic formations have been mapped in this area (Tschanz and others, 1969). Faulting, facies changes, limited exposure, and scarcity of radiometric age dates complicate the stratigraphic correlation of these strata. Sampling in the present study emphasizes formations for which

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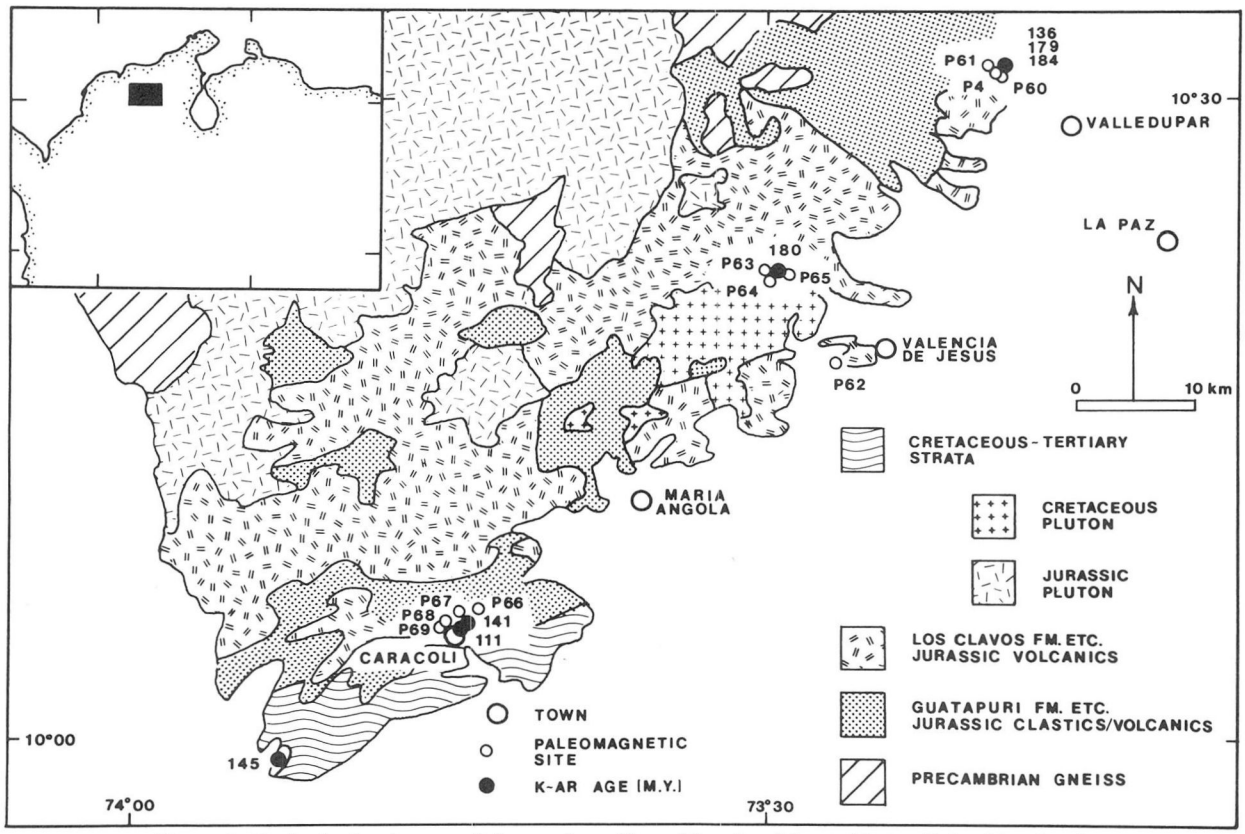


Figure 1. Geologic sketch map of the southern Sierra Nevada of Santa Marta, Colombia, showing locations of paleomagnetic sites and radiometric ages. After Tschanz and others (1969, 1974).

K-Ar radiometric ages are available. Ages noted herein have been recomputed from Tschanz and others (1974) by the method of Dalrymple (1979).

PALEOMAGNETIC SURVEY

Three rock formations were sampled: Los Clavos, Guatapuri, and Caja de Ahorros. Because only one site (P62) was sampled in the silicic volcanic Caja de Ahorros formation, and because the age of this unit is not well controlled, this formation is not discussed further. The distribution of the other units is sketched in Figure 1.

Los Clavos Formation

Rhyolitic to dacitic ignimbrites, flows, dikes, and breccias characterize this unit. Radiometric ages range mainly from 179 to 184 m.y., as recomputed from Tschanz and others (1974). Six sampling sites in two groups lie about 30 km apart near Valledupar and Valencia de Jesus (Fig. 1). The lithologies sampled include rhyolitic ignimbrite (P4, P60, P61), felsic dike (P63), granitic porphyry (P64), and epidotized agglomerate (P65). Bedding attitudes could not be discerned at sites P63, P64, and P65.

Guatapuri Formation

Red beds predominate in this formation, intercalated with silicic to basic flows. Four sites were drilled, in diverse lithologies: tuffs and flows (P66), red beds (P67), keratophyre (P68), and basalt (P69). Near Caracoli, a basalt at P69 yields a recomputed age of 141 m.y. From a nearby keratophyre (P68), we obtained a younger, presumably reset, K-Ar age of 111 ± 4 m.y. About 60 km to the northeast, near Valledupar, Guatapuri red beds rest on older basalts with recomputed ages of 165 and 181 m.y. (Tschanz and others, 1974).

Paleomagnetic Results

Paleomagnetic site mean directions are summarized in Table 1. After serial alternating field (a.f.) demagnetization of pilot samples, the remaining samples at each site were demagnetized at a single step. The a.f. demagnetizing sequence was 0, 2.5, 5, 7.5, 10, 15, and 20 millitesla (mT). Typical samples show a univectorial decay toward the origin after removal of a low coercivity component by fields below 10 mT (Fig. 2). Site P69, a scoriaceous basalt flow, showed poor clustering after a.f. demagnetization at 20 mT, and was thermally demagnetized. Six samples from site

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TABLE 1. PALEOMAGNETIC SITE MEANS FOR JURASSIC UNITS, SIERRA NEVADA DE SANTA MARTA, COLOMBIA

Site	n	Field (mT) or temperature (°C)	In situ inc.	In situ dec.	Tilt-corrected inc.	Tilt-corrected dec.	κ	α_{95}
Guatapuri Fm (10.1°N, 073.7°W; approx. age 141 m.y.)								
P66	10	25mT	71.9	289.9			3.9	28.3
P67	9	20mT	42.0	346.4	57.6	350.5	290.2	3.0
P68	8	20mT	38.4	012.8	50.8	023.0	208.3	3.9
P69	7	20mT	02.8	224.7			1.4	--
P69	4	600 °C	-26.1	184.6	-31.1	186.8	906.4	3.1
Los Clavos Fm (10.4°N, 073.4°W; approx. age 180 m.y.)								
P60	5	5mT	17.3	337.4	31.6	348.1	5.9	34.5
P61	6	15mT	17.0	342.8	29.0	353.3	19.4	15.6
P63	7	15mT	49.5	016.4			32.9	10.7
P64	6	10mT	50.3	326.9			6.0	29.7
P65	6	15mT	-26.4	046.1			6.7	27.8
Caja de Ahorros Fm (10.3°N, 073.4°W; age possibly 140-180 m.y.)								
P62	5	20mT	-35.8	340.7			1.4	--

Note: n is number of samples per site; a.f. demagnetizing field is in millitesla (mT); thermal demagnetization level is in degrees Centigrade (°C); κ is Fisher's (1953) precision parameter; α_{95} is semi-apical angle of cone of 95% confidence about site mean direction.

P69 were thermally demagnetized in the sequence 120, 210, 300, 400, 450, 500, 550, 600, 650, and 750 °C. Directions for four of the six samples so treated moved into a tight cluster, for which the Fisher (1953) statistics are reported in Table 1 at the 600 °C demagnetization level. P69 is the only reversed site detected.

Bedding tilt corrections were applied to site-mean directions for sites at which bedding was evident: P60, P61, P67, P68, and P69 (Table 1). Because the regional dip is to the southeast, with interruptions related mainly to faulting, the tilt correction increases the inclination and adds a small clockwise increment to the declination.

Formation means and paleomagnetic poles are reported in Table 2. In calculating the Los Clavos formation mean, site P65 (Table 1) was omitted. That site is an epidotized agglomerate for which the site mean direction diverges widely from other Los Clavos directions. Site P66, excluded from the Guatapuri formation mean, has scattered directions, high coercivities, and an unusual near-vertical inclination (Fig. 3). In retrospect, thermal demagnetization might possibly have been of benefit to this site.

DISCUSSION

An important issue in this preliminary study is, Do the Santa Marta results represent Jurassic paleofield directions or a younger overprint? The reversed site (P69) indicates that a paleofield is represented.

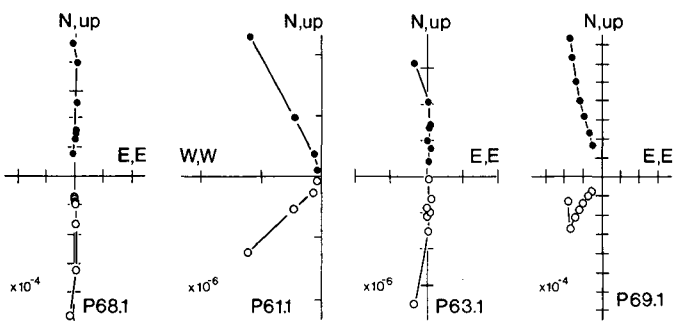


Figure 2. Typical orthogonal projections of a.f. demagnetized vectors for samples from the Jurassic of Santa Marta. Filled circles are projections into the horizontal plane, and open circles are projections into the vertical east-west plane. Units are emu/cm^3.

TABLE 2. JURASSIC PALEOMAGNETIC MEANS AND POLES

Formation	Sites	Mean direction inc.	Mean direction dec.	α_{95}	Paleomagnetic pole lat.(N)	Paleomagnetic pole long.(E)	dp	dm
Guatapuri	P67*,P68*,P69*	36.4	002.4	20.8	79.6	-061.1	14.1	24.2
	P67†,P68†,P69§	47.2	007.6	26.7				
Los Clavos	P60*,P61*,P63*,P64*	34.7	344.6	29.1				
	P60†,P61†,P63*,P64*	41.3	351.2	21.1	74.3	-104.5	15.7	25.7

*In situ, a.f. demagnetized
†Tilt-corrected, a.f. demagnetized.
§Tilt-corrected, thermally demagnetized, polarity reversed for computation of mean.

Few paleomagnetic studies of coeval rocks from cratonic South America are available for comparison. A study by Schult and Guerreiro (1979) of Jurassic basalts from the Maranhao basin of Brazil is relevant. The pole for those 158-m.y.-old basalts is at 85 °N, 083 °E, with an alpha-95 of 6.9°. The 95% confidence circle of that pole overlaps the present geographic North Pole. Diabase dikes from the Guayana shield of Venezuela have a pole at 70°N, 066°E, with dp = 2.6, dm = 5.0 (MacDonald and Opdyke, 1974). Although that pole is significantly different from the present pole, its recomputed age is 204 ± 5 m.y., near the Triassic-Jurassic boundary, estimated at 204 ± 4 m.y. (Odin and others, 1982). This diabase dike pole, therefore, probably represents a field 40 to 60 m.y. older than the Santa Marta rocks studied here.

The Santa Marta results do not indicate large tectonic rotations such as have affected the La Teta lava of the Guajira Peninsula nearby to the north (MacDonald and Opdyke, 1972). Parenthetically, we note here that the age estimate of La Teta lava has been revised to 143 ± 7 m.y., placing it near the end of the Jurassic rather than in the Cretaceous.

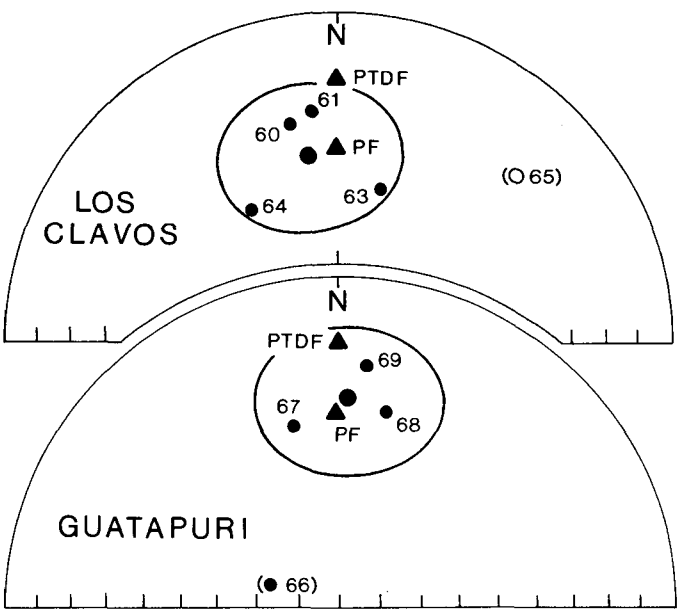


Figure 3. Site-mean directions and formation-mean directions with circles of 95% confidence for Los Clavos and Guatapuri Formations, Santa Marta massif. Site-means in parentheses have been excluded from the calculation of the formation-means. Data from Tables 1 and 2.

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Several Jurassic paleomagnetic analyses have been made elsewhere in the mobile orogenic zone peripheral to South America. A complete summary of those results is beyond the scope of this note. It is significant to observe that several research groups that have studied Jurassic rocks from that region of South America have reported paleomagnetic directions nearly parallel to those of the present theoretical dipole field (Creer and others, 1972; Vilas, 1974; Shackleton and others, 1979). Reversals are common in some of the units analyzed.

CONCLUSION

It is anticipated that distinguishing Jurassic paleomagnetic directions from younger overprints in Santa Marta, as elsewhere in South America, will be a continuing challenge. Associated ambiguities in the age of magnetization will undoubtedly complicate the development of the Jurassic polar wander path for South America, as well as the analysis and identification of allochthonous tectonic terranes in the Andes, a major task which has barely begun.

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